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Module 7 Introduction
Lesson 1: Maximum Likelihood Inference
Lesson 2: Bayesian Inference
Reading: Bayesian Inference Readings 1h
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Module 7 Summary

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item

Bayesian Inference Quiz

Review Learning Objectives

Instructions

coach
Ready to review what you've learned before starting the assignment? I'm here to help.
Help me practice Let's chat

Assignment details
Submitted: Nov 16, 11:23 PM IST
Attempts: Unlimited

Your grade
100%

View submission

1. What is a fundamental principle of Bayesian inference? 1 / 1 point

- (selected) It updates the probability estimate for a hypothesis as more evidence or information becomes available.
- It considers only the prior distribution while ignoring the observed data.
- It always results in a binary decision about the acceptance or rejection of a hypothesis.
- It relies solely on the frequency of observed data to make statistical conclusions.

Try again

Correct
Correct! Bayesian inference involves updating the probability of a hypothesis based on new data.

2. In Bayesian inference, what is a "prior"? 1 / 1 point

- A prediction about the future data to be observed.
- The decision rule used to choose between different statistical models.
- The probability of the observed data given no prior knowledge.
- (selected) The initial probability distribution representing knowledge or uncertainty about a parameter before observing data.

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Correct
Correct! A prior distribution reflects existing knowledge or beliefs before considering new evidence.

3. How does Bayesian inference differ from frequentist inference? 1 / 1 point

- Bayesian inference cannot handle large datasets, unlike frequentist methods.
- Bayesian inference is used only for predictions, while frequentist inference is used for hypothesis testing.
- (selected) Bayesian inference incorporates prior knowledge and updates beliefs with data, while frequentist inference relies on long-run frequencies without involving prior beliefs.
- Bayesian methods are purely subjective, and frequentist methods are purely objective.

Correct
Correct! The key difference lies in the use of prior knowledge and the interpretation of probability.